

Rithish S

Software Development Engineer Intern | Developer Tools, AI Agents (MCP/RAG) & Scalable Systems

[+91 99522 52304](tel:+919952252304) | rithishcodespace@gmail.com | [LinkedIn](#) | [GitHub](#) | [LeetCode](#)

October 4, 2026

Engineering & Campus Recruitment Team

HackerRank (InnovAccer / HackerRank India)

Bengaluru, Karnataka, India

Subject: Application for Software Development Engineer Intern (Jan – June 2027)

Dear HackerRank Engineering Team,

HackerRank's founding belief that technical talent should be identified and celebrated based on demonstrated skills rather than pedigree has fundamentally transformed global software hiring. Today, as software development enters the era where humans and AI build side-by-side, HackerRank's mission to invent next-generation assessment tools and empower 2,500+ world-class companies represents the exact environment where I want to push code to production. I am available full-time to work from HackerRank's Bengaluru office for the entire 6-month duration (January – June 2027).

My engineering background is deeply rooted in HackerRank's core ethos: *writing code used and trusted by other engineers around the world*. In my merged open-source contribution to **ESLint Core (PR #21218)**, I resolved an intricate edge case in the **no-loss-of-precision** AST static analysis rule distinguishing literal zeros from decimal values that underflow under IEEE 754 binary64 specifications, implementing test suites that passed **43 automated CI verification checks**. Similarly, I engineered and published **DB Backup CLI (DBVault)** on npm and Docker Hub with **120 automated tests**, integrating distributed queue processing with **Redis and BullMQ**, distributed locking, AES-256-GCM encryption, and automated AWS S3 offloading.

To build next-generation developer tooling where humans and AI collaborate seamlessly, I architected **PatentIQ**, an **autonomous AI agent** workflow using the **Model Context Protocol (MCP)** and dynamic tool calling to orchestrate vector search, document parsers, and live research endpoints. I developed a low-latency **RAG** pipeline with Pinecone vector embeddings and local **LLMs (Qwen2.5 via Ollama)**, backed by a Clean Architecture Fastify and PostgreSQL service ensuring structured, deterministic JSON execution. Backing my systems work is proven algorithmic rigor: I have solved **1200+ DSA problems** across LeetCode, CodeChef, and GeeksforGeeks, won 1st place in two collegiate hackathons (120+ and 200+ teams), and mentored **50+ students** in software engineering craft as a Web Developer for Students Special Group.

HackerRank's values—caring deeply about the craft, putting developers first, and going deep to own outcomes—align with how I approach engineering every single day. I would welcome the opportunity to learn from HackerRank's world-class mentors, contribute meaningful features to production from month one, and help build the future of developer assessment. Thank you for your time, consideration, and review of my application.

Sincerely,

Rithish S

rithishcodespace@gmail.com | +91 99522 52304